

Model Cards and Human Review

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Proxy Features and Fairness

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Overfitting

Without opening the old lesson, explain one key idea from Overfitting. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Check arithmetic for Model cards and review. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Model cards and review. Proposed explanation: “A final answer is enough”.
Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

6. Solve the given case

A model card says healthy/spotted,80% accuracy only.

Add responsible-use evidence.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A model card says 90% accuracy but gives no intended use, data conditions or human-review rule. What should be added before classroom use?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 33 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Complete the Plant Classifier model card and require teacher sign-off before any classroom demonstration.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Draft a purpose

Write an intended-use and non-use statement for the worked card.

My reasoning and evidence

13. W2 Check arithmetic

Verify the worked accuracy, precision and recall.

My reasoning and evidence

14. W3 Find missing evidence

A card says fair and reliable but provides only accuracy. Request three additions.

My reasoning and evidence

15. W4 Apply independently

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

My reasoning and evidence

16. W5 Maintain the record

What should change in a card after a threshold revision?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Model Cards and Human Review

Write a model card whose performance statement can be traced to a concrete evaluation.

Toy $v = 3$ classifier scored 16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored 14/20.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Model Cards and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name evidence

Record $v = 3$, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.



2. Compare baseline

$v = 3$ accuracy 80%; baseline 70% on the same 20 cases.



3. State limits

Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

CHANGE ONE THING

The publicity sentence becomes "80% accurate for every plant".

Explain the changed result using the matching lesson.

Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Write a model card whose performance statement can be traced to a concrete evaluation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

1. Name evidence
2. State limits
3. Compare baseline

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: State limits Do not claim all cameras, outdoor conditions or disease diagnosis were validated. Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: The publicity sentence becomes “80% accurate for every plant”. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Reject that unbounded statement.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Reject that unbounded statement. The measured rate belongs to twenty related indoor photos, not every plant or setting.

Repair a missing building block

Review Proxy Features and Fairness (ai-4/32). Goal: Identify possible proxy features. Short practice: What evidence contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Improve Generalization (ai-4/30). Goal: Compare training and validation evidence for model capacity. Short practice: What fresh evidence is needed after adding crossed training data? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Overfitting (ai-4/27). Goal: Compare training and validation evidence for model capacity. Short practice: Does 100% training accuracy prove Large was always best? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

